MATERIAL DE APOIO

Equipamento da Unimed, informação conforme manual.

2.3 Apresentação Externa

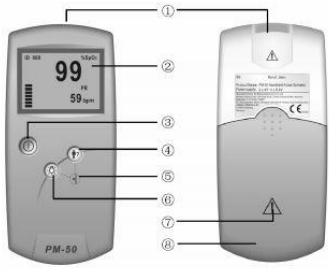


Figura 2-1 Painel Frontal Figura 2-2 Painel Preto

Tabela 2-1 Descrição da Apresentação Externa

Capítulo 4 Medição

4.1 Princípio de Medição

O Oxímetro de Pulsação PM-50 é capaz de medir SpO_2, PR e potência da pulsação.

O PM-50 mede a SpO_2 através de um método chamado oximetria da pulsação. É um método contínuo e não-invasivo com base nos diferentes espectros de absorção de hemoglobina e oxihemoglobina reduzidas. Ele mede quanta luz, enviada de fontes de luz de um lado do sensor, é transmitida através do tecido (como do dedo ou da orelha) para um receptor do outro lado. A quantia de luz transmitida é determinada por fatores múltiplos, cuja maioria é determinada. O fluxo sangüíneo nas artérias é um dos fatores que altera com o tempo, em razão da pulsação.

Geralmente, o comprimento de onda da luz transmitida por um LED de luz vermelha que pode ser detectado pelo sensor é de 660 nm, e por um LED de luz infravermelha é de 940 nm. A saída máxima de energia do LED é de 4 mW.

O PM-50 mede a luz absorvida durante o período pulsátil para adquirir a freqüência de pulsação e a potência da pulsação, bem como a saturação de oxigênio arterial e, então, após processamento, exibe os resultados na tela LCD. O valor de SpO2 exibido é o de saturação funcional.

5.1.1 Armazenamento de Dados

O PM-50 possui memória interna para armazenar dados. A memória está dividida em Zona de Dados de ID e Zona de Dados de Tendência.

A Zona de Dados de ID é capaz de armazenar, no máximo, 100 dados de ID de pacientes. Quando o número excede 100, novos dados automaticamente cobrirão o dado armazenado do mais velho ao mais novo.

Dados ID 001	Dados ID 002	Dados ID 003	•••••	Dados ID 100

Um dado de ID inclui:

- O número de valores de SpO_2 e freqüência de pulsação armazenados neste ID,
- O valor máximo de SPO2 deste ID,
- O valor mínimo de SPO2 deste ID,
- O valor médio de SPO2 deste ID,
- O valor máximo de PR deste ID,
- O valor mínimo de PR deste ID,
- O valor médio de PR deste ID.

A Zona de Dados de Tendência é capaz de armazenar, no máximo, 200 dados de tendência (TD, conforme mostrado abaixo). Quando o número excede 200, novos dados automaticamente cobrirão o dado armazenado do mais velho ao mais novo.

TD 001	TD 002	TD 003	•••••	TD 200

Os primeiros dados de tendência serão armazenados 15 segundos após o pulso ser encontrado.

Outro manual traz a mesma informação:

9.7.6 Light-Emitting Diodes (LEDs)

Two LEDs are used in Nellcor oximetry sensors. Light is transmitted through body tissue and received by a photodetector circuit that converts it to photocurrent. The two wavelengths, which are used for calculation of pulse rate and oxygen saturation in blood, are transmitted at the following frequencies:

- infrared (IR) light at approximately 915 microns
- red light at approximately 660 microns

9.7.7 Pulse Width Modulation (PWM)

The three 8-bit PWM outputs can be software controlled; their duty cycle can be changed from 0-255/256 of the total pulse duration. PWM frequency is the crystal frequency of the CPU, which is 10 MHz divided by 1024. The PWMs control the gains within the analog circuit.

9.7.8 RCal

Sensor RCal value is a resistance value specific to an individual sensor. This value is used by the software during oxygen saturation computations to maximize accuracy.

9.7.9 Real-Time Clock (RTC)

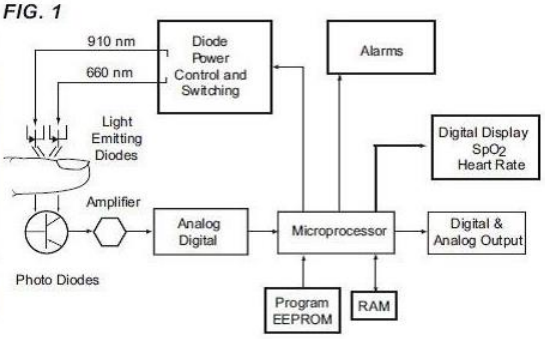
The RTC is used with the optional printer to track time and date for printouts.

Principles of Pulse Oximetry

Introduction

Pulse oximeters provide a spectrophotometric assessment of functional arterial hemoglobin oxigenation (SpO2). Pulse Oximetry is based on the following two principles: First, hemoglobin (Hb) and oxygenated hemoglobin (HbO2) differ in their absorption of red and infrared light. Second, the volume of arterial blood in tissue (and therefore light absorption by the hemoglobin) changes during the pulse. A pulse oximeter passes red and infrared light into an arteriolar bed, measures changes in light absorption, and determines SpO2.

FIG. 1



How Pulse Oximeters Work

Pulse oximeter sensors have red and infrared low voltage light emitting diodes (LEDs) wich serve as light sources. The emitted light is transmitted through the tissue, then detected by the photodetector and sent to the microprocessor of the pulse oximeter (Figure 1) All constituents of the human body, venous and arterial blood, and tissue absorb light (Figure 2). The pulsating of arterial blood results in changes in the absorption to to added hemoglobin (Hb) and oxygenated hemoglobin (HbO2) in the path of the light. Since HbO2 and Hb absorb light to varying degrees, this varying absorption is translated into plethysmographic waveforms at both red and infrared wavelengths (Figure 3).

FIG. 2

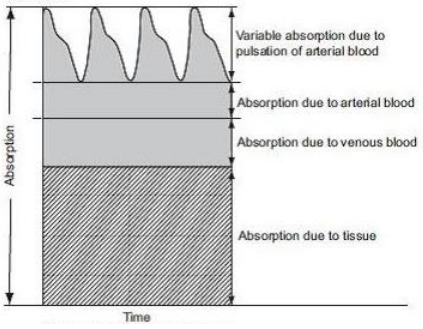


Figure 10.7 shows a fiber-optic instrument devised to measure oxygen saturation in the blood. This device, which could also be used for measuring cardiac output with a dye injected, is described here. The instrument consists of red and infrared light-emitting diodes (LEDs) and a photosensor. Plastic optical fibers are well adapted to these wavelengths. Figure 10.8 shows a fiber-optic oximeter catheter that is flow directed. After insertion, the balloon is inflated, and blood flow drags the tip through the chambers of the heart.

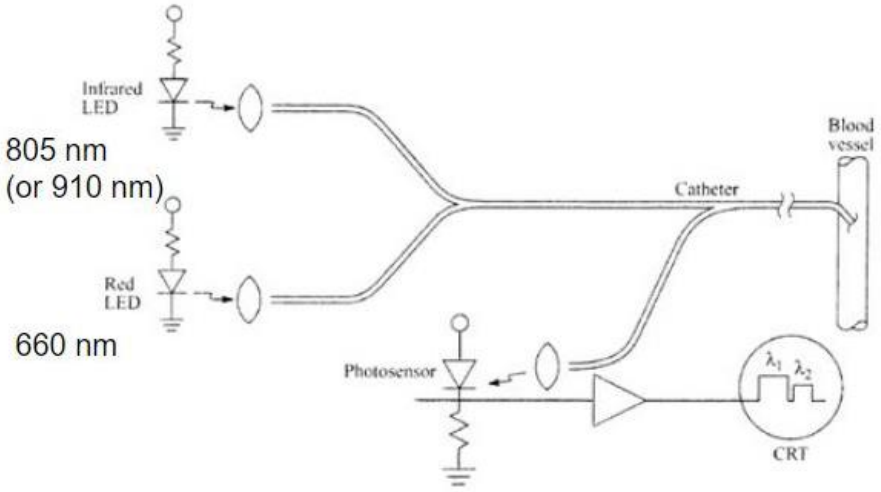


Figure 10.7 The oximeter catheter system measures oxygen saturation *in vivo*, using red and infrared light-emitting diodes (LEDs) and a photosensor. The red and infrared LEDs are alternately pulsed in order to use a single photosensor.

Oxygen saturation formula:

$$\text{Oxygen saturation} = \frac{C(HbO_2)}{C(HbO_2) + C(Hb)} \times 100\ (\%)$$

C (Hb) = Concentration of deoxygenated hemoglobin

C (HbO_2) = Concentration of oxygenated hemoglobin

Oxygen saturation is a measurement of the percentage of oxygen binding sites that contain oxygen. If all the oxygen binding sites contain oxygen, then the oxygen saturation is 100%. Oxygen saturation is defined as the ratio of oxy-hemoglobin to the total concentration of hemoglobin present in the blood (i.e. Oxy-hemoglobin + reduced hemoglobin). When arterial oxy-hemoglobin saturation is measured by an arterial blood gas it is called SaO2. When arterial oxy-hemoglobin saturation is measured non-invasively by a finger pulse oximeter or handheld pulse oximeter, it is called SpO2. Note that SaO2/SpO2 alone doesn't reveal how much oxygen is in the blood; for that we also need to know the hemoglobin content.

Absorbtivity "A" can be measured at different wavelengths, where "I" is the detector output.

$$A(\lambda) = \log\left(\frac{I_{blood}}{I_{water}}\right)$$

Beer's Law $$A(\lambda) = WL[a_o(\lambda)C_o + a_r(\lambda)C_r] \qquad (10.12)$$

where

W = weight of homoglobin per unit volume

L = optical path length

a_o and a_r = absorptivities of HbO_2 and Hb

$C_o = C_r$ = relative concentrations of HbO_2 and Hb ($C_o + C_r = 1.0$)

Figure 10.6 shows that a_o and a_r are equal at 805 nm, called the *isosbestic* wavelength. If this wavelength is λ_2, then

$$A(\lambda_2) = WLa(\lambda_2)\cdot(C_o + C_r) = WLa(\lambda_2)$$

$$WL = \frac{A(\lambda_2)}{a(\lambda_2)} \qquad (10.13)$$

where

$$a(\lambda_2) = a_o(\lambda_2) = a_r(\lambda_2) \qquad (10.14)$$

Blood Oximetry --- Based on the Absorptivities of Oxygenated Hemoglobin ($Hb0_2$) and Unoxygenated (Reduced) Hemoglobin (Hb).

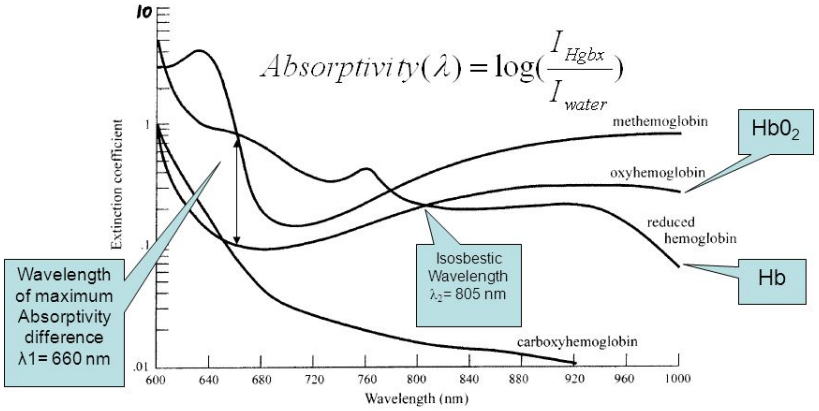


Figure 10.6 Absorptivities (extinction coefficients) in liter/(mmol·cm) of the four most common hemoglobin species at the wavelengths of interest in pulse oximetry. (Courtesy of Susan Manson, Biox/Ohmeda, Boulder, CO.)

9.6 SpO_2 Analog Circuitry Block Diagram (Figure 9-3)

Analog circuitry has high signal sensitivity and reduced susceptibility to noise. Its design allows for a wide range of input signal levels and a broad range of pulsatile modulation. The SpO_2 analog circuit (Figure 9-3) consists of four subsections:

1. Sensor output/LED control, where the CPU controls the gain of both LEDs so that signals received at the input amplifier are in its acceptable dynamic range
2. Input signal conditioning, where sensor output current is converted to voltage
3. Signal gain, where the separated LED signals are amplified so their current levels are within the A/D converter's acceptable range; and
4. AC ranging, where DC offset is eliminated from each LED signal.

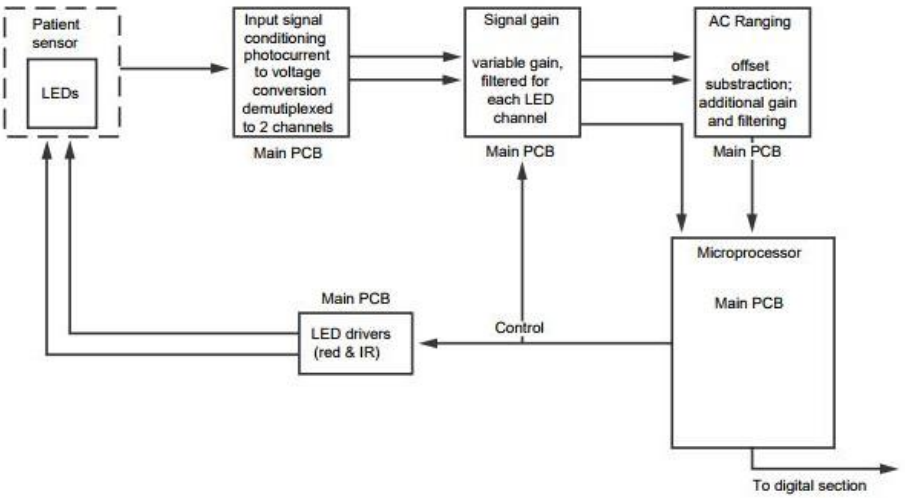
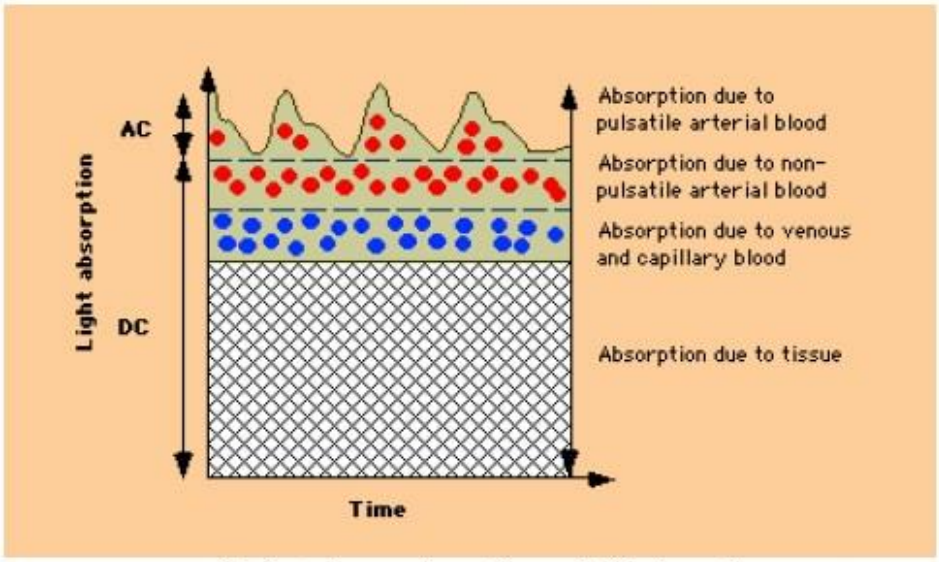


Figure 9-3: SpO_2 Analog Circuitry Block Diagram

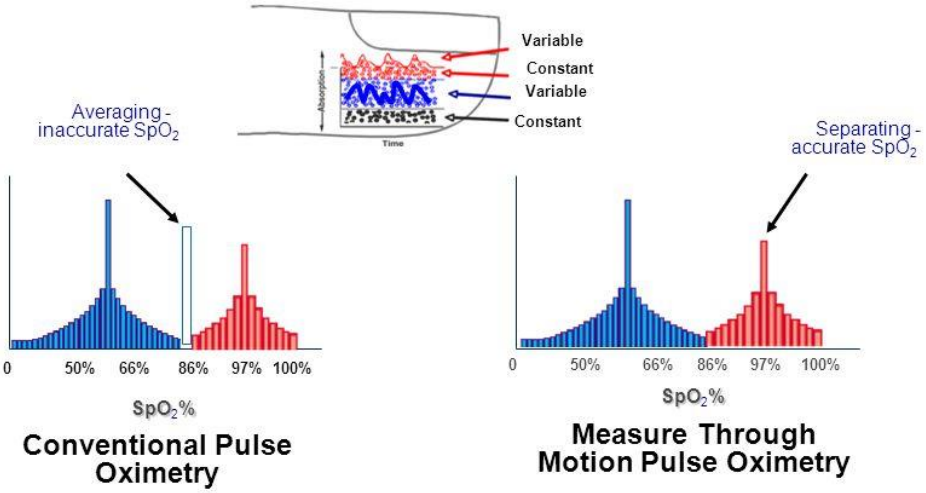
$$SpO_2 = K \times \frac{\frac{AC660nm}{DC660nm}}{\frac{AC940nm}{DC940nm}}$$



Light absorption through living tissue

A Solution for Patient Motion
Discrete Saturation Transform (DST)

In the presence of motion, SET separates the venous and arterial saturation values resulting in accurate saturation readings without false alarms (compared to conventional oximetry that averages the values to produce a reading)



Conventional Pulse Oximetry

Measure Through Motion Pulse Oximetry